

I. LAGRANGIAN FORMULATION

The dynamics of the two-wheeled inverted pendulum are derived using the Euler–Lagrange formulation. The Lagrangian is defined as

$$\mathcal{L} = T - V, \quad (1)$$

where T is the total kinetic energy and V is the gravitational potential energy.

For a system with generalized coordinates q_i , the Euler–Lagrange equations are

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i, \quad (2)$$

where Q_i is the generalized nonconservative force associated with q_i . Because the no-lateral-slip condition is nonholonomic, the equations for x and ψ require additional terms; these are derived in the equations-of-motion subsection.

A. Coordinate System

The system is described using an inertial Cartesian coordinate frame $\mathcal{F}_I$ with axes (X, Y, Z) , where Z is vertical and positive upward. A heading frame $\mathcal{F}_H$ has its origin at the midpoint of the wheel axle. Its x_H axis points forward along the robot heading, its y_H axis points to the left, and its z_H axis is parallel to Z .

The position of the axle midpoint is

$$\mathbf{r}_a = \begin{bmatrix} X \\ Y \\ z \end{bmatrix}, \quad (3)$$

where z is constant. The heading angle ψ is the rotation of $\mathcal{F}_H$ about Z , positive counterclockwise when viewed from above (a left turn). The distance traveled along the heading is x , defined by

$$\dot{x} = \dot{X} \cos \psi + \dot{Y} \sin \psi. \quad (4)$$

Note that x is a path length and not an inertial Cartesian coordinate.

The center of mass of the pendulum body is located at

$$\mathbf{r}_b = \mathbf{r}_a + \ell \begin{bmatrix} \sin \theta \cos \psi \\ \sin \theta \sin \psi \\ \cos \theta \end{bmatrix}, \quad (5)$$

where ℓ is the distance from the wheel axle to the center of mass of the pendulum body.

The pitch angle θ is measured clockwise from the positive Z axis when viewed with x_H pointing to the right, i.e., positive θ tilts the body forward. Therefore,

$$\theta = 0 \quad (6)$$

corresponds to the upright configuration of the inverted pendulum.

The generalized coordinates are

$$q_1 = x, \quad q_2 = \theta, \quad q_3 = \psi, \quad q_4 = \phi_l, \quad q_5 = \phi_r, \quad (7)$$

where ϕ_l and ϕ_r are the *absolute* (inertial-frame) angular positions of the left and right wheels, positive for forward

rolling. If wheel angles are measured relative to the body (e.g., by encoders), then $\phi = \phi^{\text{rel}} + \theta$.

The wheel angular velocities are

$$\omega_l = \dot{\phi}_l, \quad \omega_r = \dot{\phi}_r. \quad (8)$$

B. Rolling and Nonholonomic Constraints

Assuming pure rolling without slipping, the wheel velocities satisfy

$$\dot{x} = \frac{r_w}{2} (\dot{\phi}_r + \dot{\phi}_l), \quad \dot{\psi} = \frac{r_w}{w_{\text{axis}}} (\dot{\phi}_r - \dot{\phi}_l), \quad (9)$$

where r_w is the wheel radius and w_{axis} is the distance between the left and right wheels. The corresponding position constraints are

$$x - \frac{r_w}{2} (\phi_r + \phi_l) = C_x, \quad \psi - \frac{r_w}{w_{\text{axis}}} (\phi_r - \phi_l) = C_\psi, \quad (10)$$

where C_x and C_ψ are constants determined by the initial configuration.

Because the rolling constraints eliminate two degrees of freedom, ϕ_l and ϕ_r may be eliminated from the final equations of motion using

$$\dot{\phi}_{r,l} = \frac{1}{r_w} \left(\dot{x} \pm \frac{w_{\text{axis}}}{2} \dot{\psi} \right). \quad (11)$$

In addition, the wheels cannot slip sideways, so the lateral velocity u of the axle midpoint in $\mathcal{F}_H$ satisfies the nonholonomic constraint

$$u = -\dot{X} \sin \psi + \dot{Y} \cos \psi = 0. \quad (12)$$

C. Kinetic Energy

Let u temporarily denote the lateral velocity of the axle midpoint; it is set to zero after differentiation. The velocity of the pendulum-body center of mass, expressed in $\mathcal{F}_H$, has components

$$\mathbf{v}_b = \begin{bmatrix} \dot{x} + \ell \dot{\theta} \cos \theta \\ u + \ell \dot{\psi} \sin \theta \\ -\ell \dot{\theta} \sin \theta \end{bmatrix}. \quad (13)$$

Setting $u = 0$ and using $\dot{z} = 0$, the translational kinetic energy of the wheel assembly and pendulum body is

$$T_{\text{trans}} = \frac{1}{2} (m_b + m_w) \dot{x}^2, \quad (14)$$

where m_b is the mass of the pendulum body and m_w is the combined mass of both wheels. The additional translational energy of the pendulum-body center of mass due to its angular displacement is

$$T_{\text{corr}} = m_b \ell \dot{x} \dot{\theta} \cos \theta + \frac{1}{2} m_b \ell^2 \dot{\theta}^2. \quad (15)$$

The pitch rotational kinetic energy of the pendulum body is

$$T_{b,\theta} = \frac{1}{2} I_{b,2} \dot{\theta}^2. \quad (16)$$

The yaw kinetic energy of the pendulum body is

$$T_{b,\psi} = \frac{1}{2} \dot{\psi}^2 (\sin^2 \theta (I_{b,1} + m_b \ell^2) + \cos^2 \theta I_{b,3}), \quad (17)$$

where the $m_b \ell^2 \sin^2 \theta$ term arises from the lateral motion of the center of mass during yaw. The wheels do not pitch with θ , so their yaw inertia is constant:

$$T_{w,\psi} = \frac{1}{2} \dot{\psi}^2 \left(2I_{w,d} + \frac{m_w w_{\text{axis}}^2}{4} \right), \quad (18)$$

where $I_{w,d}$ is the inertia of one wheel about a diameter, and the second term is the parallel-axis contribution of the two wheels. The wheel spin energy is

$$T_{w,\text{spin}} = \frac{1}{2} I_w (\dot{\phi}_1^2 + \dot{\phi}_r^2) = \frac{I_w}{r_w^2} \dot{x}^2 + \frac{1}{2} \frac{I_w w_{\text{axis}}^2}{2r_w^2} \dot{\psi}^2, \quad (19)$$

where I_w is the spin inertia of a single wheel about its axle and (11) has been applied.

Define the effective translational mass, the effective pitch inertia, and the yaw inertia

$$m_{\text{eff}} = m_b + m_w + \frac{2I_w}{r_w^2}, \quad (20)$$

$$I_\theta = I_{b:2} + m_b \ell^2, \quad (21)$$

$$\mathcal{J}(\theta) = J_0 + \Delta I \sin^2 \theta, \quad (22)$$

with

$$\begin{aligned} J_w &= 2I_{w,d} + \frac{m_w w_{\text{axis}}^2}{4} + \frac{I_w w_{\text{axis}}^2}{2r_w^2}, \\ J_0 &= J_w + I_{b:3}, \\ \Delta I &= I_{b:1} + m_b \ell^2 - I_{b:3}. \end{aligned} \quad (23)$$

Thus $\mathcal{J}(\theta)$ equals the sum of the wheel yaw inertia and (17).

The total kinetic energy, after applying the rolling constraints, is

$$\begin{aligned} T &= T_{\text{trans}} + T_{\text{corr}} + T_{b,\theta} + T_{b,\psi} + T_{w,\psi} + T_{w,\text{spin}} \\ &= \frac{1}{2} m_{\text{eff}} \dot{x}^2 + m_b \ell \dot{x} \dot{\theta} \cos \theta \\ &\quad + \frac{1}{2} I_\theta \dot{\theta}^2 + \frac{1}{2} \mathcal{J}(\theta) \dot{\psi}^2. \end{aligned} \quad (24)$$

When $u \neq 0$ is retained, the only additional term that survives in the derivatives evaluated at $u=0$ comes from the lateral kinetic energy $\frac{1}{2} m_b (u + \ell \dot{\psi} \sin \theta)^2$ of the pendulum body, giving

$$\left. \frac{\partial T}{\partial u} \right|_{u=0} = m_b \ell \dot{\psi} \sin \theta. \quad (25)$$

D. Potential Energy

The gravitational potential energy of the pendulum body is

$$V = m_b g Z_b = m_b g (z + \ell \cos \theta). \quad (26)$$

Since z is constant, the term $m_b g z$ is constant and does not affect the equations of motion. It can therefore be omitted, yielding

$$V = m_b g \ell \cos \theta. \quad (27)$$

E. Constrained Lagrangian

Using (24) and (27), the constrained Lagrangian is

$$\begin{aligned} \mathcal{L} &= \frac{1}{2} m_{\text{eff}} \dot{x}^2 + m_b \ell \dot{x} \dot{\theta} \cos \theta \\ &\quad + \frac{1}{2} I_\theta \dot{\theta}^2 + \frac{1}{2} \mathcal{J}(\theta) \dot{\psi}^2 - m_b g \ell \cos \theta. \end{aligned} \quad (28)$$

F. Generalized Forces

Let M_{lw} and M_{rw} denote the torques produced by the left and right motors. Each motor applies a torque M to the pendulum body in the positive θ direction and an equal and opposite torque $-M$ to its wheel. The virtual work is

$$\delta W = (M_{lw} + M_{rw}) \delta \theta - M_{lw} \delta \phi_l - M_{rw} \delta \phi_r. \quad (29)$$

Substituting the constraints $\delta \phi_{r,l} = (\delta x \pm \frac{w_{\text{axis}}}{2} \delta \psi) / r_w$ gives the generalized forces

$$Q_\theta = M_{lw} + M_{rw}, \quad (30)$$

$$Q_x = -\frac{M_{lw} + M_{rw}}{r_w}, \quad (31)$$

$$Q_\psi = \frac{w_{\text{axis}}}{2r_w} (M_{lw} - M_{rw}). \quad (32)$$

Positive M tilts the body forward ($+\theta$) and drives the vehicle backward. A larger right-wheel speed turns the robot to the left, which is the positive ψ direction. The yaw torque sign follows from $\dot{\psi} = r_w (\dot{\phi}_r - \dot{\phi}_l) / w_{\text{axis}}$.

G. Equations of Motion

Because x and ψ are subject to the nonholonomic constraint (12), the equations for these two coordinates are obtained from the Hamel (quasi-velocity) form of Lagrange's equations,

$$\left. \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} - \dot{\psi} \frac{\partial \mathcal{L}}{\partial u} \right|_{u=0} = Q_x, \quad (33)$$

$$\left. \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\psi}} + \dot{x} \frac{\partial \mathcal{L}}{\partial u} \right|_{u=0} = Q_\psi, \quad (34)$$

while θ is an ordinary holonomic coordinate and uses (2) directly.

1) *Translational Equation:* The momentum conjugate to $\dot{x}$ is

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m_{\text{eff}} \dot{x} + m_b \ell \dot{\theta} \cos \theta. \quad (35)$$

Taking the time derivative gives

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = m_{\text{eff}} \ddot{x} + m_b \ell (\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta). \quad (36)$$

Using (25) in (33), the translational equation of motion is

$$\begin{aligned} m_{\text{eff}} \ddot{x} + m_b \ell \cos \theta \ddot{\theta} - m_b \ell \sin \theta (\dot{\theta}^2 + \dot{\psi}^2) \\ = -\frac{M_{lw} + M_{rw}}{r_w}. \end{aligned} \quad (37)$$

The $\dot{\psi}^2$ term is the centripetal term arising from the lateral offset of the center of mass while turning.

2) *Pendulum Angular Equation:* Applying the Euler-Lagrange equation to θ gives

$$\frac{\partial \mathcal{L}}{\partial \theta} = m_b \ell \dot{x} \cos \theta + I_\theta \ddot{\theta}. \quad (38)$$

Taking the time derivative gives

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = m_b \ell (\ddot{x} \cos \theta - \dot{x} \dot{\theta} \sin \theta) + I_\theta \ddot{\theta}. \quad (39)$$

The partial derivative with respect to θ is

$$\frac{\partial \mathcal{L}}{\partial \theta} = -m_b \ell \dot{x} \dot{\theta} \sin \theta + \frac{1}{2} \dot{\psi}^2 \Delta I \sin 2\theta + m_b g \ell \sin \theta. \quad (40)$$

Therefore, the pitch (θ) equation of motion is

$$I_\theta \ddot{\theta} + m_b \ell \cos \theta \ddot{x} - \frac{1}{2} \dot{\psi}^2 \Delta I \sin 2\theta - m_b g \ell \sin \theta = M_{lw} + M_{rw}. \quad (41)$$

The first term represents the rotational inertia of the pendulum body, the second represents translational–rotational coupling, the third represents yaw-induced torque, and the fourth represents the gravitational torque.

3) *Yaw Equation of Motion:* The momentum conjugate to $\dot{\psi}$ is

$$\frac{\partial \mathcal{L}}{\partial \dot{\psi}} = \mathcal{J}(\theta) \dot{\psi}. \quad (42)$$

Taking the time derivative and using $d\mathcal{J}/dt = \Delta I \sin 2\theta \dot{\theta}$ gives

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = \mathcal{J}(\theta) \ddot{\psi} + \Delta I \sin 2\theta \dot{\theta} \dot{\psi}. \quad (43)$$

Using (25) in (34), the yaw equation of motion is

$$\mathcal{J}(\theta) \ddot{\psi} + \Delta I \sin 2\theta \dot{\theta} \dot{\psi} + m_b \ell \sin \theta \dot{x} \dot{\psi} = \frac{w_{\text{axis}}}{2r_w} (M_{lw} - M_{rw}). \quad (44)$$

The second term is the gyroscopic coupling between pitch rate and yaw rate, and the third term comes from the nonholonomic constraint.

H. Symbol and Coordinate Definitions

TABLE I
SYMBOLS AND COORDINATE DEFINITIONS.

Symbol	Definition
$\mathcal{F}_I$	Inertial Cartesian frame with axes (X, Y, Z) ; Z positive upward.
$\mathcal{F}_H$	Heading frame at the axle midpoint: x_H forward, y_H left, z_H vertical.
X, Y	Inertial horizontal position of the axle midpoint.
z	Height of the axle above the ground; constant.
x	Distance traveled along the heading, $\dot{x} = \dot{X} \cos \psi + \dot{Y} \sin \psi$ (path length, not an inertial coordinate).
u	Lateral velocity of the axle midpoint in $\mathcal{F}_H$; constrained to zero (no lateral slip), used only to form the nonholonomic terms.
ψ	Heading (yaw) angle about Z , positive counterclockwise viewed from above (left turn).
θ	Pitch angle of the pendulum body from the positive vertical axis, positive forward; $\theta = 0$ is upright.
ϕ_l, ϕ_r	Absolute (inertial-frame) angular positions of the left and right wheels, positive for forward rolling.
ω_l, ω_r	Wheel angular velocities, $\omega_{l,r} = \dot{\phi}_{l,r}$.
$\mathbf{r}_a$	Position vector of the axle midpoint.
$\mathbf{r}_b$	Position vector of the pendulum-body center of mass.
Z_b	Vertical coordinate of the pendulum-body center of mass.
$\mathbf{v}_b$	Velocity of the pendulum-body center of mass in $\mathcal{F}_H$.
ℓ	Distance from the wheel axle to the pendulum-body center of mass.
r_w	Wheel radius.
w_{axis}	Distance between the left and right wheels (track width).
m_b	Mass of the pendulum body.
m_w	Combined mass of both wheels (each wheel center of mass on the axle).
$I_{b:1}, I_{b:2}, I_{b:3}$	Principal moments of inertia of the pendulum body about its center of mass: axis 1 forward, axis 2 the pitch axis (perpendicular to the plane of motion), axis 3 along the body's upright axis.
I_w	Spin moment of inertia of a single wheel about its axle.
$I_{w,d}$	Moment of inertia of a single wheel about a diameter (used for yaw).
m_{eff}	Effective translational mass, $m_b + m_w + 2I_w/r_w^2$.
I_θ	Effective pitch inertia, $I_{b:2} + m_b \ell^2$.
J_w	Constant wheel yaw inertia, $2I_{w,d} + m_w w_{\text{axis}}^2/4 + I_w w_{\text{axis}}^2/(2r_w^2)$.
J_0	Yaw inertia at $\theta = 0$, $J_w + I_{b:3}$.
ΔI	$I_{b:1} + m_b \ell^2 - I_{b:3}$.
$\mathcal{J}(\theta)$	Total yaw inertia, $J_0 + \Delta I \sin^2 \theta$.
g	Gravitational acceleration.
T	Total kinetic energy of the system.
V	Gravitational potential energy of the system.
$\mathcal{L}$	Lagrangian, $\mathcal{L} = T - V$.
q_i	i th generalized coordinate.
Q_i	Generalized nonconservative force associated with q_i .
M_{lw}, M_{rw}	Left and right motor torques; each acts on the body in the positive θ direction and on its wheel in the opposite direction.
Q_x, Q_θ, Q_ψ	Generalized forces for x , θ , and ψ from the motor torques.
C_x, C_ψ	Constants determined by the initial rolling configuration.
t	Time.